Phase A FINAL FROZEN — Change Log

Corrections authorized by Section B.1 of `Phase_A_Source_Audit.pdf`

No.	Location	Exact change
1	**§4.2, equation (10)**	Replaced the malformed literal "le" by $\leq$ in $(\varepsilon X)^d \Delta_K^{-1/2} \leq \lvert P\rvert \leq (2\varepsilon X+1)^d$.
2	**§4.2, additive-advantage box**	Replaced $\lvert P\rvert \gg X^d \Delta_K^{-1/2}$ by $\lvert P\rvert \geq (\varepsilon X)^d \Delta_K^{-1/2}$, retaining the dimension-dependent factor ε^d.
3	**§3, Lemma 3.1 input**	Added the exact hypothesis that K is totally real of degree $d \geq 2$ to $R_K \leq \Delta_K$.
4	**§8, sumset normalization**	Replaced the attribution to directness by the actual inclusion: $1 \in G$ implies $P = 1 \cdot P \subseteq A$, hence $\lvert A\rvert \geq \lvert P\rvert$.
5	**§4.2 and §9 parameter table**	Distinguished the formal Lemma 4.1 hypotheses $X, Y \geq 2$ from the proof reduction to sufficiently large X, Y and integral X, with bounded ranges and rounding absorbed into constants.
6	**§1–§10 labels and ledger**	Marked conceptual summary boxes INTERPRETATION; in the gain-versus-cost table, marked displayed estimates PAPER and qualitative mechanism/gain readings INTERPRETATION.
7	**§11.1, Lemma 4.1 → Theorem 1.2**	Inserted the fixed rescaling and rounding: for theorem parameter $x \geq 1$, choose integral $X_L \asymp x$ with a fixed proportionality absorbing $Y, D_0^{3/2}$, and the Lemma 4.1 constant; then rename x as X.
8	**§12, number-field supply**	Added the citation to BSSZ §5 for the statement that the final sum-product proof does not use the earlier split-prime, class-group-fibre, or bounded-denominator construction.
9	**§13, unit-distance table**	Recast the class-group/fixed-denominator versus regulator/discriminant row as a contrast between different cost ledgers, not a direct correspondence.
10	**§13, projection row**	Added that the injective projections do not lose cardinality, while the unit-distance proof separately loses at most a factor two when directed pairs become unordered edges.
11	**§14, mechanism table**	Replaced $e^{-sd} = \lvert A\rvert^{-\delta}$ by the correct implication $\lvert A\rvert \leq e^{Md} \Longrightarrow e^{-sd} \leq \lvert A\rvert^{-s/M}$.
12	**Conceptual summary boxes**	Added INTERPRETATION labels to previously unlabelled conceptual summaries, including the central answer, high-degree explanation, component roles, gain/cost competition, comparison summary, and recall chain.

No proof architecture or source-verified core estimate was changed. No Phase B material was added.